



BT302 IMMUNOLOGY

SHORT NOTES (MID & FINAL)



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FINAL-TERM

BT302 (IMMUNOLOGY)

CHAPTER # 21

What is immunodeficiency?

The deficiency or failure of immune system is called as immunodeficiency. It is a state of complete absence of immunity or effective immune response. There are defects regarding generation of immune reactivity against infectious agents like bacteria & virus etc. There is no immune response against transformed cells like tumours or malignancy.

Who are immunodeficient patients?

The persons having immunodeficiency are also called as immunocompromised patients because there is increased vulnerability for opportunistic infections in addition to other forms of infections. As there is decrease surveillance against tumours so more chances of getting tumours are there in immunodeficient patients.

How many categories are classified of Immunodeficiency?

Immunodeficiency is classified into two main categories based on mode of acquiring

1. Primary Immunodeficiency
2. Secondary Immunodeficiency

Explain the Classification of Immunodeficiency?

Primary Immunodeficiency

Primary immunodeficiency is basically the inherited or congenital defects of the immune system. These defects can be either in specific or non-specific immune systems. Primary immunodeficiency diseases can occur in individuals of any age. These are classified based on the site of lesion in the developmental or differential pathway of immune system. The chances of susceptibility to variety of infections get increased in this kind of immunodeficiency. The nature of infection depends on the kind of immunodeficiency. There are more than 400 primary immunodeficiencies. Some are relatively common, while others are quite rare. Some affect a single cell or protein of the immune system and others may affect two or more components of the immune system.

Secondary Immunodeficiency

This kind of immunodeficiency is not congenital in nature in contrast to primary immunodeficiency. This state of immunodeficiency is associated with infections for example AIDS. There is also association of aging factor like hypo-cellularity in bone marrow. Malignancies for example Leukaemia and Myeloma are also linked with secondary immunodeficiency. Furthermore, other metabolic disorder like diabetes and renal malfunction etc. are also linked with secondary immunodeficiency. The most common secondary immune deficiencies are caused by aging, malnutrition, certain medications and some infections, such as HIV.

Write down the kinds of Primary immunodeficiency?

Primary immunodeficiency can be of two kinds

Immunodeficiency in specific immune system

Those conditions due to defects of the adaptive immune system in which defence is carried out in a more specific manner by T-cells and antibody producing B-cells. This kind of immunodeficiency is due to defects in stem cell differentiation.

Immunodeficiency in Non-Specific immune system

Those fewer common conditions with defects in the innate immune system, a system of cells and mechanisms that defend the host from infection in a non-specific manner.

Write down the important disease of immunodeficiency in specific immune system in

Primary Immunodeficiency?

Followings are the important disease of immunodeficiency in specific immune system

Reticular dysgenesis

It is either complete absence or severe deficiency of lymphocytes & granulocytes in blood due to decrease differentiation of bone marrow stem cells.

Severe combined immunodeficiency (SCID)

This disorder is due to the involvement of lymphoid stem cells. There is complete absence of T & B cell immunity. The patients suffering from SCID are susceptible to variety of bacterial, viral, mycotic & protozoan infections.

DiGeorge Syndrome

This is the disorder of T-lymphocytes due to congenital thymic aplasia or hypoplasia. This condition is highly immunodeficient as live vaccines also cause infections in the patients suffering from DiGeorge syndrome.

X-linked hypo gammaglobulinemia

This kind of immunodeficiency is due to functional disorder of B-Lymphocytes. Such patients

are highly susceptible to recurrent bacterial infections.

IgA deficiency

There is deficiency of IgA in this disorder as a result individuals are susceptible to mucosal surface infections like GIT, eye & nasopharynx.

Selective IgG deficiency

In this kind of immunodeficiency, there is selective deficiency of IgG which leads towards hyper IgM immunodeficiency.

Write down the important disease of immunodeficiency in Non-Specific immune system in Primary Immunodeficiency?

Followings are the important conditions which are linked with immunodeficiency in non-specific immune system

Cyclic Neutropenia

There is low number of circulating neutrophils in the blood which cause recurrent bacterial infections among such patients.

Chronic Granulomatous disease (CGD)

This kind of immunodeficiency is due to defects in phagocyte functions.

Complement deficiency

There is deficiency of complement proteins in this kind of immunodeficiency. Patients suffering from such disorder are susceptible to infections particularly Neisseria.

What is AIDS?

Acquired Immunodeficiency Syndrome (AIDS). This condition is caused by a virus called as human immunodeficiency virus (HIV). This condition is initiated through an initial infection with influenza like illness. Further, the disease progress with other defects in immune system leading towards severe immunodeficiency. The patients get severe opportunistic infections like tuberculosis. Moreover, the chance of getting tumour also increased. AIDS is linked with abnormalities in circular lymphocytes. AIDS patients have also increased susceptibility to various following tumours

- Kaposi sarcoma
- Burkitt's lymphoma
- Primary central nervous system lymphoma

Write down the mode of transmission of AIDS?

For AIDS, HIV infection is transmitted by following two modes

Horizontal transmission

This kind of transmission occurs usually by person to person contact either sexually or through body fluids like blood and semen.

Vertical transmission

This form of transmission is from mother to fetus during the process of pregnancy, delivery & breast feeding.

What do you know about T-cells deficiency with variable degree of B-cells deficiency?

This form of immunodeficiency is linked with T-lymphocytes deficiency which also affects the number of B-lymphocytes with a variable degree. In this condition, both cell-mediated as well as humoral immunity get affected leading to multiple forms of infections.

Write note on Immunodeficiency disorders of T-Cells?

As T-lymphocytes have a very significant role in providing cell-mediated and humoral immunity, that's why immunodeficiency associated with T-cells disorders affects both cell-mediated & humoral immunity. There is complete absence & functional abnormality in T-cells which increase the chances of getting multiple infections like viral, protozoal & fungal infections. In immunodeficiency associated with T-cells the incidence of viral infections like cytomegalovirus and measles gets increased.

DiGeorge Syndrome

It is one of the primary immunodeficiencies in which there is complete absence of T-lymphocytes. This condition is also called as thymic aplasia or hypoplasia. This form of immunodeficiency is also linked with hypoparathyroidism due to abnormal development of fetus. There is poor development of heart, thymus & parathyroid due to which there is abnormal development of heart, thymus & parathyroid. All patients suffer from DiGeorge syndrome have thymic aplasia. This disease is autosomal dominant caused by deletion in chromosome number 22. This deletion is of variable size which doesn't correlate with severity of disease. The treatment of DiGeorge syndrome occurs through thymic graft only.

T-cells deficiency with variable degree of B-cells deficiency

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Ataxia-telangiectasia

This form of T-cells associated immunodeficiency has deficient T-cells with less movement of blood vessels which also affects the function of T-lymphocytes.

Wiskott-Aldrich Syndrome

In this condition, there is normal number of T-cells with reduced functions as a result chances of getting multiple infections become increased.

Write note on Immunodeficiency disorders of B- Cells?

This form immunodeficiency is associated with normal T-cells; however, number of B-lymphocytes may be low or normal. As a result, the serum Immunoglobulin levels are low. There are more chances of getting pyogenic infections like bacterial infections.

X-linked hypo-gammaglobulinemia

In this form of B-cells associated immunodeficiency, the numbers of B-cells are very low. Due to which the immunoglobulin levels are also very low. This condition is linked with defect in B-cell maturation. Patients suffer from this immunodeficiency are prone to recurrent bacterial infections.

IgA deficiency

This is the commonest form of immunoglobulin deficiency. There is defect in class switching phenomenon of immunoglobulin synthesis. These patients are prone to mucosal surface infections like GIT, eye & nasopharyngeal infections. This form of immunodeficiency is diagnosed through measurement of serum IgA levels by various immunological methods

X-linked hyper IgM -Immunodeficiency

This form of B-cell associated immunodeficiency has very high levels of IgM. As a result, the levels of IgG & IgA get reduced. The underlying defect in this condition is in class switching due to defect in CD40 ligand on CD4 T-cells. The patients are susceptible to pyogenic infections like bacterial infections. This condition can be treated with intravenous gamma globulins injections.

Write down the note on Immunodeficiency defects of Phagocytic Cells?

The primary immunodeficiency associated with cells of non-specific immune system including phagocytic cells like neutrophils, monocytes, and macrophages. There is also defect in killer cells like NK cells.

Congenital Agranulomatosis

This form of congenital immunodeficiency has decreased neutrophil count. There is defect in myeloid progenitor differentiation into neutrophils. These patients are prone to pyogenic infections like bacterial.

Chronic Granulomatous disease (CGD)

This form of primary immunodeficiency has decreased neutrophil function while their number remains normal. The underlying defect is due to poor intracellular killing ability of neutrophils. This defect of neutrophils is due to deficiency of NADPH oxidase & other co-factors which are required for respiratory burst. CGD patients have increased susceptibility to bacterial infections.

Leukocyte Adhesion deficiency

As the name suggests this form of immunodeficiency is linked with defect in integrin molecules as a result there is decreased process of diapedesis & defective neutrophils movement towards chemotactic signals. This defective phagocytic function leading towards recurrent bacterial infections among such patients.

Write down the note on Immunodeficiency defects of Complement System?

In this form of immunodeficiency of non-specific immune system, there are abnormalities in complement proteins which are also called as Hypo-complementemia. These defects are genetic in nature due to inherited defects in the synthesis of various complement proteins. There is major defect in the synthesis of C3 due to defective C3 synthase. Moreover, defects are also due to various regulatory proteins like Factor H & I. Majority of complement deficiency is autosomal recessive in nature except properdin deficiency which has X-linked inheritance. However, MBL deficiency can be both. The patients suffering from such disorders have recurrent bacterial infections particularly Neisseria & Streptococcus infections. Furthermore, these conditions also cause autoimmune disorders like SLE, vasculitis etc.

CHAPTER # 20

Define tumor?

An abnormal mass of tissue that forms when cells grow and divide more than they should or do not die when they should. Tumors may be benign (not cancer) or malignant (cancer). Benign tumors may grow large but do not spread into, or invade, nearby tissues or other parts of the body.

Write Use of Tumor Neo-antigens in Immuno-diagnosis?

Tumor neo-antigens are used for In-vivo detection of relatively small tumour foci (masses) through radiolabelled monoclonal antibodies against tumor antigens. Likewise, the In-vitro use for finding the cell origin of undifferentiated tumour particularly lymphocytic origin in the blood & bone-marrow. Furthermore, Immuno-histochemical use of monoclonal antibodies against metastatic foci of tumours can help in the diagnosis of such tumours.

Write use of Neo-antigen in immunotherapy?

The tumour neo-antigens can also be used for Immunotherapy for treating various tumours in different following ways

Active immunotherapy

In this form of immunotherapy, host actively participates against tumors. Active immunotherapy is achieved by inducing immune response by injecting irradiated tumour cells having neo-antigens on them. The active form of immunotherapy has following two forms.

a) **Non-specific active immunotherapy**

This is generated for activation of immune response against neo-antigens using various vaccines E.g., BCG for activation of macrophages against tumours.

b) **Specific active immunotherapy**

In this form, the killed tumor cells or their antigens are used for killing of tumor cells.

Passive immunotherapy

This form of immune therapy uses the pre-formed antibodies against specific neoantigens of tumor through following ways. The specific monoclonal antibodies can be used against tumour antigen can be served for

- a) As a vehicle for delivering anti-cancer drug to tumour or cancer.
- b) Activation of components of innate immune system like complement proteins.

Explain Escape of Tumor from Immuno-Surveillance?

The mechanisms which is responsible for tumor rejection in host is termed as Immunosurveillance mechanisms. Tumour cells develop strategies for evading immune surveillance Tumours escape from immune response by following mechanisms

- I. Tumor cells may not express neo-antigens which are immunogenic in nature as a result no immune response is generated against such tumours.
- II. Tumor cells may fail to express co-stimulatory molecules for activation of T-lymphocytes for generating effective cell mediated immune response against tumours.
- III. Certain tumours may lack or poor expression of MHC for proper immune response generation.
- IV. At early development of tumor, there is low level of antigens which are in sufficient for activation of immune system which is termed as low dose tolerance.
- V. Overwhelming of immune system, after sudden maturation & expression of neo-antigens on tumor cells which is called as high dose tolerance.
- VI. Certain tumours secrete immunosuppressive molecules for inactivation of immune cells.
- VII. Some tumours shed their antigens for neutralization of antibodies.

What is Tumor Associated Antigens?

To generate effective immune response by immune system, tumor must have certain antigens on its surface. These antigens are generated through alteration in number of gene expression by a process called as tumorigenesis. The expression of new antigens on the surface of tumor are called as neo-antigens. Moreover, there is also alteration in existing antigens which are present on normal cells. These antigens are membrane bounded receptors, regulators of cell cycle & apoptosis and molecules of signal transduction.

Write two main types of tumours associated antigens?

Tumor specific transplantation antigens

These antigens are unique to tumor cells. These are not expressed on normal cells. These antigens are responsible for rejection of tumor by host immune system. In most cases, these antigens cannot be easily identified.

Tumor Associated transplantation antigens

These antigens are expressed by both tumor & normal cells in contrast to tumor specific transplantation antigens. Various chemicals, UV radiations & viruses are responsible for expression of these neo-antigens. Majority of these tumours are weakly immunogenic or on immunogenic.

What is Tumour Associated Transplantation Antigens? Or What are onco-fetal antigens?

The tumor antigens which are also expressed by normal cells. These are expressed in high levels by tumour cells as compared to normal cells. These kinds of antigens are also called as onco-fetal antigens as these are expressed during early development & lost during adult life. However, these antigens re-expressed on tumour cells. These antigens also serve as tumor markers due to their importance in the diagnosis & prognosis of cancers.

What are the types of onco-fetal antigens?

Alpha-fetoproteins (AFP)

These antigens are found as secretory protein in serum. Their level gets raised during hepatocellular carcinoma.

Carcino-embryonic antigens (CEA)

These antigens are found both in secretory & cell associated form. Their level gets raised in colon cancer.

What is the immune response against Tumours?

The immune response to tumours is complex. Cells of the immune system can inhibit tumour growth and progression through the recognition and rejection of malignant cells, a process referred to as immunoediting.

Does the immune system fight tumours?

The immune system can help to fight cancer. Some cells of the immune system can recognise cancer cells as abnormal and kill them. But this may not be enough to get rid of a cancer altogether. Some treatments aim to use the immune system to fight cancer.

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Difference between Benign and Malignant Tumours?

Benign Tumours

Benign tumours do not spread from origin to other parts of the body.

Malignant Tumours

Malignant tumours can spread to other parts of the body from the point of origin.

CHAPTER # 19

What is the process of transplantation?

The process of moving cells, tissues, and organs from one site to another among individuals is termed as transplantation. In this process of transplantation, one person called as donor who donates tissue to another person termed as recipient.

Define donor and recipient?

One person called as donor who donates tissue to another person termed as recipient.

Write kinds of tissue transplantation based on the source of donated tissue?

Isograft

This is a form of tissue which is transplanted between individuals of same genetic makeup as identical twins.

Autograft

It may be from the same person tissue to another site within same individual.

Allograft

This form of transplantation which can be from one person to other non-identical person.

Xenograft

This kind of transplantation is across the species for example from animal to human.

Write down the role of Immune system in tissue transplantation?

Immune system plays an important role in tissue transplantation as it acts as a barrier for transplantation. Immune system identifies tissue transplant as foreign and mount an immune response leading destruction or damage of transplanted tissue. This process through which recipient's immune system activation occur against donor's tissue is termed as transplantation rejection. The immunocompetent host recognize foreign antigens on grafted tissues as a result immune response is generated against grafted tissue is called as host vs graft rejection. Similarly, tissue transplants in immunocompromised individuals lead to immune response by immunocompetent cells in graft is termed as graft vs host rejection. In this case, host antigens serve as foreign. However, if there is no immune response against donated tissue by the recipient's immune system due to similar genetic makeup then it is called as graft acceptance.

Define transplantation rejection?

The process through which recipient's immune system activation occur against donor's tissue is termed as transplantation rejection.

Write down the kinds of tissue rejection?

Host vs. Graft Rejection

In this kind of transplant rejection, there are antigens on the surface of graft which are being recognized as foreign by host immune system.

Graft vs. Host Rejection

In this form of transplant rejection, the lymphoid (immune) tissues in graft recognize host immune system as foreign as a result immune response is mounted.

Write the Immune Mechanisms of Graft Rejection?

The reaction of host against allo-antigens of graft which is also called as host vs graft (HVG) rejection. This considered as a main obstacle in organ transplantation. The involvement of immune mechanisms in graft rejection is based on

- Time of rejection
- Nature of allo-antigens of graft
- Immune status of host

Define graft acceptance?

If there is no immune response against donated tissue by the recipient's immune system due to similar genetic makeup, then it is called as graft acceptance.

What are transplantation antigens?

The transplantation antigens which are also termed as Major Histocompatibility Complex (MHC) or

Human Leucocytes Antigens (HLA). The MHC exist as a complex which is encoded by group of genes on a same chromosome called as haplotype.

Write down the types of MHC?

Class I MHC

These genes are for Human MHC which are located at chromosome no 6. These contain following three major loci

1. Locus B
2. Locus C
3. Locus A

Each major locus encodes for polypeptide for MHC structure. In which there is a α -chain that contains antigenic determinants.

Class II MHC:

The class II MHC is encoded by genes for Human MHC which are located at chromosome 6. Structurally, the Class II MHC complex is also composed of three following major loci.

1. DP
2. DQ
3. DR

Each of these loci code for one alpha & one beta chain. Both of alpha and beta chains associate together to form class II MHC. Likewise, to class I, class II antigens are also polymorphic in nature which means these contain many alleles of α -chains. The DR locus contain more than one beta chain genes and possibly four in various combinations. The Class II MHC is expressed on B-lymphocytes and antigen presenting cells (APC).

Write short note on according to the time of rejection what are types of graft rejection?

Hyper-acute Rejection

This kind of transplant rejection has very quick onset for tissue rejection. It occurs within minutes to hours. The high titer of pre-formed antibodies against antigens of graft is responsible for such form of reactions. Antigen/antibody (immune) reaction occurs on the tissue surface. Afterward the fixation of complement leads towards graft destruction.

Accelerated Rejection

This kind of rejection is also called as secondary or 2nd set rejection. It occurs after transplantation of second graft. It is due to sharing of antigenic determinants with the first transplant. Time required for this form of rejection is within 2-5 days. The sensitized T-cells during first graft are responsible for this kind of rejection. Moreover, the Lymphokines and cytotoxic T-lymphocytes (CTLs) augment it.

Acute Rejection

It is also called as primary, or 1st set of tissue rejection. It occurs during first graft with allo-antigen on grafted tissue. The time span required for such form of rejection is 1-3 weeks. These reactions are mediated by sensitized T-cells to class I & II of allo-graft. Furthermore, the secretion of Lymphokines leads

towards the activation of monocyte and macrophages.

Chronic Rejection

This form of transplantation rejection is called as delayed rejection which occur within months to years. After transplantation the graft remains normal for months to years, but sudden rejection occurs due to still unknown mechanisms. These are many hypotheses reading such form of rejection like the involvement of various infections which are responsible for loss of immunological tolerance by grafted tissue in host.

Write the prevention and treatment of Graft rejection?

If there is decrease in tissue rejection then there is relative increase in survival of graft. There are successful grafts occur for example mostly in case of kidney & cornea. For successful transplantation there is need of better understanding of immune response and MHC. The success of tissue graft is based on following important factors

Donor Selection

For appropriate donor selection, there should be MHC compatibility with recipient. Identical twins are considered as the ideal donor which are also termed as Isograft. HLA matched siblings have 95-100% chance of graft success. One haplotype of parent or sibling must be HLA-D matched for successful graft. Moreover, ABO compatibility is also essential for high success rate of transplant

Recipient Preparation

For successful tissue graft, the recipient should be in good health with no current infection. Similarly, the absence of active malignancy would also increase the success rate. Moreover, in recipient the absence of any systemic diseases would lead towards better rehabilitation. Likewise, the recipient should not be hypertensive. One to five transfusions of 100-200 ml of donor's blood at 1-2 weeks interval would also increase the compatibility level with donor.

Immunosuppression

This is the way through which host immune system can be suppressed and is most essential component of allo-transplantation. For this there is usage of following immunosuppressive drugs with their specific mode of actions.

Cyclosporin A

It inhibit IL-2 synthesis following antigen exposure to host as a result no immune response is generated for allo-antigens.

Rapamycin

It inhibits signal transduction for the activation and differentiation of immune cells. For instance, inhibition of T-cells proliferation and activation is achieved by this.

What uses of transplantation of Blood Cells & Bone Marrow?

Transplantation of blood and bone marrow is used for treating various haematological disorders. The success of these graft is achieved by having good compatibility between donor and recipient. For this purpose, following procedures are used

- ABO blood grouping
- Cross matching
- HLA typing

Difference between Vaccination and Immunization?

Vaccination: The act of introducing a vaccine into the body to produce protection from a specific disease. Vaccination is the administration of a vaccine to help the immune system develop protection from a disease. Vaccines contain a microorganism or virus in a weakened or killed state, or proteins or toxins from the organism.

Immunization: A process by which a person becomes protected against a disease through vaccination. Immunization is the process whereby a person is made immune or resistant to an infectious disease, typically by the administration of a vaccine.

Vaccination	Immunization
The process involves introducing a weakened / deactivated disease causing microbes into a person	The process starts after the person is exposed to the vaccine and the body starts building resistance to that disease
It is usually injected or administered orally	It is not administered in any way. The body develops resistance from vaccines.
Imovax Rabies is the trade name for rabies vaccine	The body builds up immunity through this vaccine for the disease rabies.
Vaccination does not guarantee complete resistance to a disease	Complete immunity occurs when the person fully recovers from the disease.
Usually, if mutation happens to microbes, it might render the vaccine ineffective (this is the reason why common cold has no vaccine)	Similarly, variations of a disease impact the body's ability to generate an immune response.

Write down the role of Adjuvants in Immunization?

An adjuvant is an ingredient used in some vaccines that helps create a stronger immune response in people receiving the vaccine. In other words, adjuvants help vaccines work better.

Adjuvants: substances which increase the antigenicity of weak antigens. It can be of two forms

Chemical adjuvants:

- Aluminium salt (Alum)
- Only chemical suitable for human use
- Used in DTP (Diphtheria, Tetanus & Pertussis) Role of Adjuvants in Immunization
- Alum causes slow release of antigen
- Increase TLR interaction
- Increase the activation of mononuclear phagocytes
- Also induce the increased cytokine secretion

Appropriate immune response generation Role of Adjuvants in Immunization

Biological adjuvants:

- Certain bacteria used as biological adjuvant
- Bacterial products are used in combination with oil & detergent
- Activation of mononuclear phagocytes & cytokine secretion

What is Immunology?

The study of all aspects of the immune system, including its structure and function, disorders of the immune system, blood banking, immunization, and organ transplantation.

Example:

Individual inflammatory cells adhering to endothelial cells in vitro is an immunological response.

Write the adverse Effects of Vaccination?

Adverse effects: Live microbe

Most severe effects are rare, mild & moderate

Active immunization: causes fever, malaise & discomfort

Joint pains (Arthritis): Rubella vaccination

Can be fatal: in case of Pertussis

Neurological disorders: Influenza vaccination Adverse Effects of Vaccination DPT vaccination local effects: Redness, swelling & pain

Mild/Moderate effects: fever, drowsiness. Vomiting & anorexia

Severe effects: persistent crying, fever, Convulsions, collapse, acute encephalopathy, permanent neurological deficit.

Define Tolerance?

Tolerance is defined as the lack of specific immune response to antigen. There is a central and peripheral tolerance of B or T cells. Central tolerance of T cells, which makes the repertoire of T cells in the periphery non-responsive to self-antigens, is clonal deletion in the thymus (or negative selection).

Write down the example of tolerance?

Self-tolerance: Physiologically no immune response against self-antigens.

Autoimmunity: immune response against self-antigens (Pathology) • Immune system recognize self-antigens & mount strong immune response • Discrimination B/w Self & Non-self-antigens: Self MHC recognition

Modification of antigens:

- Leads towards tolerance
- Most bacteria & viruses develop tolerance: exploit or evading host immune system
- Lepromatous type of leprosy: Tolerance against M. leprae

What is the tolerance of tissues?

Tolerance to tissue & cell antigens can be induced artificially by injecting haemopoietin stem cells in neonatal or severely immunocompromised animals. Chimeras' animals: transferring of allogeneic primary lymphoid tissues at early life and induction of tolerance against allogeneic tissues

How can tolerance break?

Tolerance can be break in following two ways

Naturally: In case of autoimmune disorders e.g., Rheumatoid Arthritis, SLE etc.

Artificially: exposure to immunosuppressive drugs or X-ray irradiation e.g., in case of experimental animals for bone marrow transplantation

Define mechanism of tolerance induction?

Two forms of immunological tolerance

Central Tolerance: occur in primary lymphoid organs e.g., bone marrow & thymus.

Peripheral tolerance: occur at secondary lymphoid organs e.g., spleen, lymph nodes, tonsils etc.

Define Central Tolerance?

Central tolerance (also known as negative selection) is the process of eliminating any developing T or B lymphocytes that are reactive to self. Through elimination of autoreactive lymphocytes, tolerance ensures that the immune system does not attack self-peptides.

Define Peripheral Tolerance?

Peripheral tolerance is the second branch of immunological tolerance, after central tolerance. It takes place in the immune periphery (after T and B cells egress from primary

lymphoid organs). Its main purpose is to ensure that self-reactive T and B cells which escaped central tolerance do not cause autoimmune disease.

Write about the tolerance to Tissues & Cells?

Tolerance is defined as the lack of specific immune response to antigen. There is a central and peripheral tolerance of B or T cells. Central tolerance of T cells, which makes the repertoire of T cells in the periphery non-responsive to self-antigens, is clonal deletion in the thymus (or negative selection). Specific unresponsiveness against various cells & tissues • Inhibition of immune response against antigens of cells & tissues. These antigens are foreign to immune system (Allogeneic antigens).

Occur in tissue graft: Allogenic graft Tolerance to Tissues & Cells.

Induction of Tolerance against tissues & cells: by injecting hematopoietic stem cells at neonatal stage

Early stage: no development of fully matured lymphoid tissues.

Tolerance induction: by immunosuppression e.g., lethal irradiation for killing of host's own stem cells in primary organs Tolerance to Tissues & Cells.

Also, **by using immunosuppressive drugs:** to inactivate immune cells of host

Chimeras: animals with hybrid nature of immune cells with host's own cells & donor's cells. Both transplantation of donor's bone marrow & thymus in early age or by immunosuppression Functional inactivation

Define Autoimmunity?

Autoimmunity is the system of immune responses of an organism against its own healthy cells, tissues, and other body normal constituents. Any disease resulting from this type of immune response is termed an "autoimmune disease". Both antibodies & T-cells are involved in autoimmunity.

What is Genetic predisposition?

Certain genes of Immunoglobulins, TCR & MHC are associated with various autoimmune diseases.

Environmental factors: responsible for autoimmunity e.g., drug & infection induced autoimmune diseases

What is Aetiology of Autoimmunity?

Exact mechanism is still unknown but various theories have been proposed for understanding the mechanism of autoimmunity. Sequestered antigens are the escape of auto reactive cellular clones or lack of regulatory T-cells and cross-reactive antigens aetiology of Autoimmunity Sequestered antigens.

What are Sequestered antigens?

Lymphoid cells may not be exposed to certain self-antigens during differentiation. Certain self-antigens are confined to specialized organs (Testis, brain & eyes etc.) and release of such antigens from tissues due to any injury or accident and Initiation of autoimmune diseases

How auto reactive clones escaped?

- Loss of central tolerance
- Auto-reactive T-cells escape from thymus to periphery
- Not all self-antigens are presented to T-cells in thymus

What is the lack of regulatory T-cells?

Few regulatory T-cells in autoimmune diseases. Absence of regulatory (Suppressive) T-cells and absence of immunosuppressive cytokines like TGF- β & IL10. Absence of Regulatory T Cells Causes Phenotypic and Functional Switch in Murine Peritoneal Macrophages. Tissue macrophages are important components of tissue homeostasis and inflammatory pathologies.

Write down the classification of Autoimmunity?

Autoimmunity classification based on tissues or organs involved Two following categories

- Organ-specific autoimmunity
- Non-organ Specific autoimmunity

Organ Specific autoimmunity		
Disease	Organs	Antibody to
Hashimoto's thyroiditis	Thyroid	Thyroglobulin, thyroid peroxidase (microsomal)
Primary Myxedema	Thyroid	Cytoplasmic TSH receptors
All hemolytic anemia	RBC	RBC antigens
Good Pasteur's Syndrome	Kidney, Lung	Renal & Lung basement membrane
Ulcerative colitis	Colon	Colon Lipopolysaccharide

Non-Organ Specific autoimmunity		
Disease	Organs	Antibody to
Rheumatoid Arthritis	Skin, kidney, Joints etc	IgG
Systemic Lupus Erythematosus (SLE)	Skin, joints	DNA, RNA, Nucleoproteins
Sjogren's syndrome	Moister-producing glands	Basement membrane
Scleroderma	Skin, blood vessels, muscles & internal organs	DNA, RNA, Nucleoproteins
Sarcodosis	Lung, skin, hear, nervous system	Auto reactive T-cells

Write down the diagnosis of Autoimmune diseases?

Diagnosis of autoimmune diseases is based on symptoms & detection of autoantibodies. Autoantibodies can be against two kinds of antigens

- Self-cell or cell associated antigens
- Soluble antigens
- Biochemical test

Autoantibodies against self-cell/Cell associated antigens using tissues section like kidney, skin etc.

By Immunofluorescence:

- Detection of autoantibodies.
- Linear Pattern (Goodpasture Syndrome)
- Granular Pattern (Systemic Lupus Erythramatous)

Define Goodpasture Syndrome?

Good pasture syndrome is a group of acute illnesses that affects the lungs and kidneys. It involves an autoimmune disorder. Normally, the immune system makes antibodies to fight off germs.

Define Systemic Lupus Erythramatous?

Systemic lupus erythematosus (SLE) is the most common type of lupus. SLE is an autoimmune disease in which the immune system attacks its own tissues, causing widespread inflammation and tissue damage in the affected organs. It can affect the joints, skin, brain, lungs, kidneys, and blood vessels.

What are Autoantibodies against Soluble Antigens?

Using serum for autoantibodies by following techniques

- Agglutination e.g., Anti-nuclear antibodies (ANA) for SLE & RA, ELISA e.g., Rheumatoid factor for Rheumatoid Arthritis.
- RIA for detection of ant thyroid antibodies

Discuss Biochemical Assays?

A "Biochemical Assay" is an analytical procedure to detect and quantify cellular processes (e.g., apoptosis, cell signalling) or metabolic reactions. Biochemical assays are a reliable, routinely used procedure that helps in characterizing targets and understanding of bimolecular functions.

- For Intrinsic factor (IF) in case of Pernicious anaemia
- Competition for TSH receptors in case of Graves' disease
- Auto-reactive T-cells can also be detected by Flow cytometry
- Treatment of Autoimmune diseases
- Treatment for autoimmune disorders
- Anti-inflammatory drugs – to reduce inflammation and pain.
- Corticosteroids – to reduce inflammation.
- Pain-killing medication – such as paracetamol and codeine.
- Immunosuppressant drugs – to inhibit the activity of the immune system.
- Physical therapy – to encourage mobility.

Mainly in research

- Anti-TNF α receptor antibodies
- Anti-IL-2 receptor antibodies
- Anti CD4 antibodies
- Anti TCR antibodies
- Anti-Duo type antibodies against autoantibodies

What do you mean by intracellular killing?

During inflammation, ingested bacteria are killed by phagocytes. After phagocytosis the ingested bacteria are being killed by a process called as Intracellular Killing.

How many ways of intracellular killing also write their name?

There are two ways of intra cellular killing:

- Oxygen independent
- Oxygen dependent

What is the mechanism of oxygen independent killing?

No need of oxygen for such kind of intracellular killing of bacteria. Granules & Vesicles of phagocytes secrete hydrolytic proteins. Those proteins are bactericidal in nature according to their modes of action mechanisms of Oxygen Independent killing.

Write a note on oxygen dependent intracellular killing?

The oxygen compounds are toxic to both the invader and the cell itself, so they are kept in compartments inside the cell. This method of killing invading microbes by using the reactive oxygen-containing molecules is referred to as oxygen-dependent intracellular killing.

Requirement of oxygen for such kind of intracellular killing of bacteria also called "Respiratory Burst" as requirement of glucose & oxygen increased after phagocytosis. Oxygen containing bactericidal radicals are produced.

What is the role of cytokines in inflammation?

- Cytokines: Mediators of Inflammation
- Complex variety of mediators in acute inflammatory response.
- Some cytokines directly act on smooth muscles wall surrounding the arterioles: to alter the blood flow. Others act on venules to cause contraction of endothelial cells opening of junctions
- Migration of leucocytes from bloodstream
- Up regulate the expression of adherence molecules on endothelial cells also adherence molecules on the surface of Leucocytes.
- Cytokines also lead the leucocytes towards the inflamed site through chemotaxis

How many types of inflammatory cytokines?

There are two types of inflammatory cytokines that are as follows:

Pro-inflammatory cytokines

Pro-Inflammatory cytokines: involved in up regulation of inflammatory process and predominantly produced by activated macrophages & TH cells. They include IL-1 β , IL-6 & TNF- α , IL-12 & INF- γ . The role in mediating innate immune response. Inflammatory diseases like atherosclerosis, cancer & depression.

Anti-inflammatory cytokines

Anti-Inflammatory cytokines involved in down regulation of inflammatory process. Control pro inflammatory cytokine response. It includes **IL-4, IL-10, IL-11& IL-13**. Pathological role in systemic inflammatory states and balance between Pro & anti-inflammatory cytokines

What is meant by Inflammasomes in inflammation?

Inflammasome is a multiprotein complex mainly functioning as a platform for the activation of inflammatory caspases which then lead to the maturation of pro inflammatory cytokines such as interleukin-1 β (IL-1 β) and interleukin-18 (IL-18). Inflammasomes expressed in myeloid cells. It is a multi-protein oligomer and responsible for activation of inflammatory responses

What is edema?

Edema is swelling caused by excess fluid trapped in your body's tissues. Although edema can affect any part of your body, you may notice it more in your hands, arms, feet, ankles, and legs.

What are three main processes that are involved in acute inflammation?

Three main processes are as follows:

- Increased blood flow
- Increased permeability
- Migration of Leucocytes

Difference between acute inflammation and chronic inflammation?

Inflammation:

Inflammation is part of the protective response of the body tissues to adverse stimuli, like irritants, pathogens, or damaged cells. It involves immune cells, molecular mediators, and blood vessels.

Acute Inflammation: Acute inflammation is the early (short term) response of the body to adverse stimuli.

Chronic Inflammation: The chronic inflammation is an inflammatory reaction that lasts for months or years.

What are the adverse effects of inflammation?

Inflammation is associated with general flu like symptoms including:

- Fever
- Chills
- Fatigue/Lethargy/Loss of energy
- Headaches
- Loss of appetite

- Body pains
- Muscle stiffness

What is vaccination?

A vaccine is a biological preparation that provides active acquired immunity to a particular infectious disease. A vaccine typically contains an agent that resembles a disease-causing microorganism and is often made from weakened or killed forms of the microbe, its toxins, or one of its surface proteins.

Write a note on the types of vaccination?

Two modes of providing specific protection against many common & damaging pathogens. These modes of immunization can be natural or artificial in their nature.

Active Immunization

In active immunization the individual's own immune system is stimulated to produce antibodies and lymphocytes.

Examples:

Active immunization stimulates the immune system to produce antibodies against a particular infectious agent. Active immunity can arise naturally, as when someone is exposed to a pathogen. For example, an individual who recovers from a Covid-19 is now immune to further infection.

Passive immunization:

In passive immunization a person receives antibodies or lymphocytes that have been produced by another individual's immune system.

Examples:

Passive immunity can occur such as

- **Naturally** when an infant receives a mother's antibodies through the placenta or breast milk.
- **Artificially** when a person receives antibodies in the form of an injection (gamma globulin injection).

What are cellular barriers which provides innate immunity in humans?

- Neutrophils
- Macrophages
- Natural Killer (NK) Cells

- Eosinophils

For which acute infections we use artificial passive immunity?

- Diphtheria
- Tetanus
- Measles
- Rabies

Artificial passive immunity is also used against various poisoning conditions like

- Insects or sting biting e.g., immunization against various strains of Malaria producing Plasmodium
- Reptiles e.g., anti-venom administration in case of snake biting
- Food poisoning (Botulism)

Vaccines which are used as artificial active immunization?

- Live (attenuated) organisms
- Killed organisms
- Sub-unit vaccines

What is artificial active immunization?

Artificial active immunization is where the microbe, or parts of it, are injected into the person before they can take it in naturally.

Differentiate between live attenuated organisms, killed organisms and sub-unit vaccine?

Live-attenuated vaccines

These vaccines use a weaker virus which can grow and replicate but does not cause disease. E.g.: Yellow fever, Measles.

Inactivated vaccines

These contain viruses whose genetic material has been destroyed by heat, chemicals, or radiation so they cannot infect cells and replicate but can still trigger an immune response. E.g.: Hepatitis A, Flu.

Subunit, recombinant, conjugate, and polysaccharide vaccines

These vaccines use certain parts of the bacteria or virus that cause a strong immune response in the body because they use a specific part of the virus. E.g.: Hepatitis B

What do you mean by novel vaccine?

Novel ways of designing vaccines

- Reduction in toxicity
- Can provoke both humoral & cell-mediated immunity
- Mainly used in experimentally
- Would be available for clinical use in future

Define the Immunodeficiency?

Immunodeficiency also known as immuno-compromisation is a state in which the immune system's ability to fight infectious diseases and cancer is compromised or entirely absent.

What is compromised immune system?

When your immune system is weakened, your body can't fight off viruses, bacteria, or fungi very well. This can lead to serious infections and can pose a major health risk. Many different types of immune system disorders can cause health problems.

What is an example of an immunodeficiency disease?

Examples of secondary immunodeficiency disorders include AIDS. cancers of the immune system, like leukaemia. immune-complex diseases, like viral hepatitis.

What are 3 possible reasons for immunodeficiency?

The most common causes worldwide include malnutrition, poor sanitary conditions, and human immune deficiency virus (HIV) infection.

Difference between types of immunodeficiency?

Types of Immunodeficiency

Primary immunodeficiency

- Born with defect
- Genetic mutation in an immune system protein

Secondary immunodeficiency

- Born with normal immune system

Caused by other factors

- Malnutrition
- Viruses (HIV)
- Irradiation/chemotherapy

- Corticosteroids
- Leukaemia
- Metabolic disease (e.g.- diabetes, liver disease)

Transient immunodeficiency

Characterized by:

- Recurrent infections
- Reduced serum Igs
- Patients typically record at around **30-40 months** of life.

Difference between Immunodeficiency and immunosuppression?

Immunosuppression

Immunosuppression generally refers to both beneficial and potential adverse effects of decreasing the function of the immune system.

Immunodeficiency

Immunodeficiency generally refers solely to the adverse effect of increased risk for infection.

Write down the disorders of T-cells?

Immunodeficiency associated with T-cells disorders effects both cells mediated & humoral immunity. Complete absence & functional abnormality in T-cells and more chances of getting viral, protozoal & fungal infections. Viral infections like cytomegalovirus virus & measles.

What is DiGeorge Syndrome?

Primary immunodeficiency is the complete absence of T-cells also called as thymic aplasia or hypoplasia. Immunodeficiency linked with hypoparathyroidism due to abnormal development of fetus and poor development of heart, thymus & parathyroid

Why all the patient has thymic aplasia?

Due to abnormal development of fetus and poor development of heart, thymus & parathyroid that's why patients have thymic aplasia.

Who Autosomal dominant caused?

- Autosomal dominant caused by deletion in chromosome 22.
- Deletion is of variable size doesn't correlate with severity of disease

- **Treatment:** Thymic graft
- T-cells deficiency with variable degree of B-cells deficiency

Define Phagocytic Cells?

Phagocytes are cells that protect the body by ingesting harmful foreign particles, bacteria, and dead or dying cells.

What are two types of phagocytic cells?

In humans, and in vertebrates generally, the most-effective phagocytic cells are two kinds of white blood cells: the macrophages (large phagocytic cells) and the neutrophils (a type of granulocyte).

Which immune cells are phagocytic?

Macrophages, neutrophils, monocytes, dendritic cells, and osteoclasts are among these dedicated cells. Professional phagocytes are responsible of removing microorganisms and of presenting antigens to lymphocytes to activate an adaptive immune response.

What are the 3 types of phagocytic cells?

The three main types of phagocytes are **neutrophils, monocytes, and macrophages**. They destroy microbes by phagocytosis. They form the cellular barrier and are responsible for innate immunity.

Define Defects of Phagocytic Cells?

Primary immunodeficiency associated with cells of non-specific immune system including Phagocytic cells like neutrophils, monocytes & macrophages also killer cells like NK cells.

Define Congenital A granulomatosis in Defects of phagocytic Cells?

- Decreased neutrophil count
- Defect in myeloid progenitor differentiation into neutrophils
- Patients are prone to pyogenic infections like bacterial

Chronic Granulomatous disease (CGD) In defect of phagocytic cells?

- Decreased neutrophil function with normal number
- Defect is due to poor intracellular killing ability of neutrophils
- Deficiency of NADPH oxidase & other cofactors for respiratory burst
- Susceptibility to bacterial infections

What are the leukocyte Adhesion Deficiency in Defects of Phagocytic Cell?

- Defect in integrin molecules.
- Decreased process of diapedesis & defective neutrophils movement towards chemotactic signals.
- Defective phagocytic function leading towards recurrent bacterial infections

Define Immunology?

Immunology focuses on the human body's built-in defence system. In a healthy person, the immune system helps the body fight infection by rejecting foreign viruses and bacteria. When the immune system is defective, it can fail to protect the body, or even attack it.

Difference between t cells and b cells in the immune responses?

An important difference between T-cells and B-cells is that B-cells can connect to antigens right on the surface of the invading virus or bacteria. This is different from T-cells, which can only connect to virus antigens on the outside of infected cells. Your body has up to 10 billion different B-cells.

Define Complement System?

The complement system, also known as complement cascade, is a part of the immune system that enhances (complements) the ability of antibodies and phagocytic cells to clear microbes and damaged cells from an organism, promote inflammation, and attack the pathogen's cell membrane.

What are the 3 main functions of the complement system?

At the basic level the broad functions of the complement system can be split into three areas:

1. The activation of inflammation
2. The opsonization (labelling) of pathogens and cells for clearance/destruction
3. The direct killing of target cells/microbes by lysis.

What are the components of complement system?

These are covered by nine central components of the cascade (**C1 to C9**), multiple activation products (such as **C3a and C3b**), regulators and inhibitors (e.g., **Factor H and C4BP**), proteases and newly assembled enzymes (e.g., **C4b2a and Factor B**), or effector molecule receptors (such as **C3aR and C5aR**).

Write about some defects of Complement System?

- Abnormalities in complement proteins (Hypo-complementarian) are genetic abnormalities
- Inherited defects in the synthesis of various complement proteins
- Major defect in the synthesis of C3 due to defective C3 synthase also defects in various regulatory proteins like Factor H & I Defects of Complement System
- Majority of complement deficiency is autosomal recessive
- Properdin deficiency: XLinked inheritance
- MBL deficiency can be both
- Recurrent bacterial infections particularly Neisseria & Streptococcus infections • Also cause autoimmune disorders like SLE, vasculitis etc.

Write a short note about Cytokines?

Cytokines belong to diverse group of non-antibody proteins which act as mediators between cells. Initially, these proteins were considered as products of immune cells. More importantly, these were also considered to act as mediators for immune cells. Later, it was also found that these are also produced from non-immune cells and served as mediators too. Cytokines are used as biological response modifier for treatment of various diseases. In general, cytokine is a general term, but it can be more specific term which is used according to their cell of origin as follows

- **Monokines**

These are produced by mononuclear phagocytic cells.

- **Lymphokines**

These are produced by activated lymphocytes.

- **Interleukins**

These are the mediators between leucocytes

- **Chemokines**

These are responsible for leucocytes migration and inflammatory process

Cytokines usually perform their functions as cascade signalling in the form of a network which is termed as cytokine network. Their functions are for enhancing their effects of other cytokines in an additive or synergistic manner. Moreover, these act for suppressing the effects of other cytokines in an opposing or antagonistic way. Cytokines are not as preformed proteins like antibodies. These are produced by gene transcription as needed because mRNA for cytokines is short lived-in nature. Furthermore, the individual cytokine can act on many cells of the host which is termed as pleiotropic response of cytokines.

What is the mechanism of acting of Cytokines?

Cytokines act through specific receptors present on various cells. These are considered as redundant in nature. The redundancy of cytokines is due to the nature of cytokine receptors. The complex interaction between host's cells and cytokines occurs through cytokine network. Different cells can respond to same cytokine due to structural similarities among cytokine receptors. The cytokine signalling is flexible which can have both protective and damaging effects. Similarly, this can also influence the synthesis of other cytokines.

How cytokines bind to specific receptors?

Cytokines bind to specific receptors with high affinity and respond in following three ways

- **Autocrine**

These effect on same cells which secrete the cytokine

- **Paracrine**

In this way, the cytokines effect on nearby cells of cytokine secreted cells.

- **Endocrine**

Through this way, the cytokines effect on distant cells through circulation.

What are the categories of Cytokines?

1. Cytokines are categorized on following two bases
2. Categories based on source cells or their origin

Followings are the categories of cytokines **based on their source of origin**

- **Monokines**

As the name indicates, these are produced by mononuclear phagocytic cells for example Interferon gamma and Interleukin (IL-1) from macrophages & Monocytes.

- **Lymphokines**

These are produced by activated lymphocytes for example IL-3, IL-4 & IL-5.

- **Interleukins**

These are mediators between leucocytes for example IL-1, IL-10& IL-18.

- **Chemokines**

These are responsible for leucocytes migration for example IL-8.

Based on functions

Based on functions, cytokines are categorized into following two categories

- Cytokines which act for Innate (non-Specific) immune system e.g. TNF- α , IL-1, IL-10 etc.
- Cytokines which act for Adaptive (Specific) immune system e.g. IL-2, IL-4 & IL-5

Write down the cytokines of Innate Immune System?

These cytokines which act primarily for innate (non-Specific) immune system. These are as follows

Tumor Necrosis Factor-Alpha (TNF- α)

It is produced by activated macrophages in response to microbes or their products for example lipopolysaccharides (LPS) of gram-negative bacteria. It acts a mediator of acute inflammation. It recruits neutrophils and macrophages at the site of infection either by stimulating endothelial cells to secrete adhesion proteins or by secreting chemokines. Moreover, TNF- α induces fever and acute phase proteins in host.

- IL-1

This is also an inflammatory cytokine as like TNF- α . It is produced by activated macrophages. Furthermore, it also helps to activate T-cells.

- IL-10

It is an inhibitory cytokine which inhibits the cytokine production of cytokines from macrophages.

- IL-12

This cytokine is produced by dendritic cells and macrophages. It enhances the cytolytic activity of cytotoxic T-cells

Type I interferon

This class of interferon include Interferon α & β . These inhibit viral replication in cells by increasing the expression of class I MHC which in turn make them susceptible to cytotoxic T-lymphocytes (CTL).

- IFN- γ

It is produced primarily by Th1 cells. It enhances the cytolytic activity of NK cells by induction of class I & II MHC molecules.

- **Chemokines**

These are chemotactic cytokines produced by many leucocytes. These recruits' inflammatory cells at site of infection for an inflammatory response.

Write down the cytokines of Adaptive Immune System?

Followings are the cytokines which act primarily for Adaptive (Specific) immune system

- **IL-2**

It is produced by T-helper (Th) cells, but Cytotoxic T-cells (Tc) also produced it in lesser extent. It stimulates B-cell division. It can also activate NK cells and monocytes. Moreover, IL-2 acts on T-cells in autocrine manner. Activation of T-cells result in expression of IL-2R receptors and production of IL-2. IL-2 binds to IL-2R and promotes cell division. However, IL-2 receptor decay if T-cells are not activated and eventually proliferative phase ends.

- **IL-4**

IL-4 is produced by macrophages and Th2 cells. It promotes the development of Th2 from naïve T-helper (Th) cells. The Differentiated Th2 cells result in the production of antibodies. IL-4 also involves in the class switching of antibody into IgE.

- **IL-5**

IL-5 is produced by Th2 cells. It promotes the development of B-cells and eosinophils. It also activates mature eosinophils.

Tumor Growth Factor (TGF)- β

TGF- β is produced by T cells and many other cell types. It acts as inhibitory cytokine which inhibits the proliferation of T-cells and activation of macrophages. It also acts on neutrophils and endothelial cells to block the effects of pro-inflammatory cytokines.

What is cytokine network?

A complex series of overlapping and inter-related connections among cytokines. As cytokines are secreted from one kind of cells and have effect on other type of cells & organs. Within this network, some cytokines have synergistic effects on other cytokines while some have antagonistic effects. Through this network, lymphocytes, macrophages, and other tissues are regulated for their functions. Similarly, monocytes, lymphocytes and other components are also communicated through this network.

How Immuno-Regulation by Cytokine?

The control of immune response between lymphocytes and macrophages. The balance is required between antigen driven activation of lymphocytes and regulatory influences which is called as immune regulation. Immuno-regulation occurs at following three phases of immune responses

- Recognition phase
- Activation Phase
- Effector Phase

The cytokines are considered as positive or negative regulator of immune response. The cytokines can act on many stages of immune responses. Their activity is dependent on presence of other cytokines in the tissue's microenvironment. It also depends upon receptor expression on effector cells. This kind of regulation regulates the nature and extent of generated immune response.

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⇒ Immunoregulation by cytokines
It has 3 phases

↓ Recognition phase ↓ It means foreign material e.g. virus, bacteria etc.	↓ Activation phase ↓ Humoral immune response has an A.P., helper T cells becomes activated.	↓ Effector phase ↓ Short lived activated cells that depends the body
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⇒ Cytokines positive or negative regulator
⇒ Cytokines can act on many stage of IR.
⇒ Working depends on presence of other cytokines.
⇒ Regulate type & extent of IR (Immune Response).

Immune Evasions
Tumour cells evade host immune's system.
Immune surveillance:
↓
Protection against pathogens
⇒ Increase ability of organism to develop infections.
⇒ Occur by weak immune response.
⇒ CMI immunity can also be evaded by microbes.

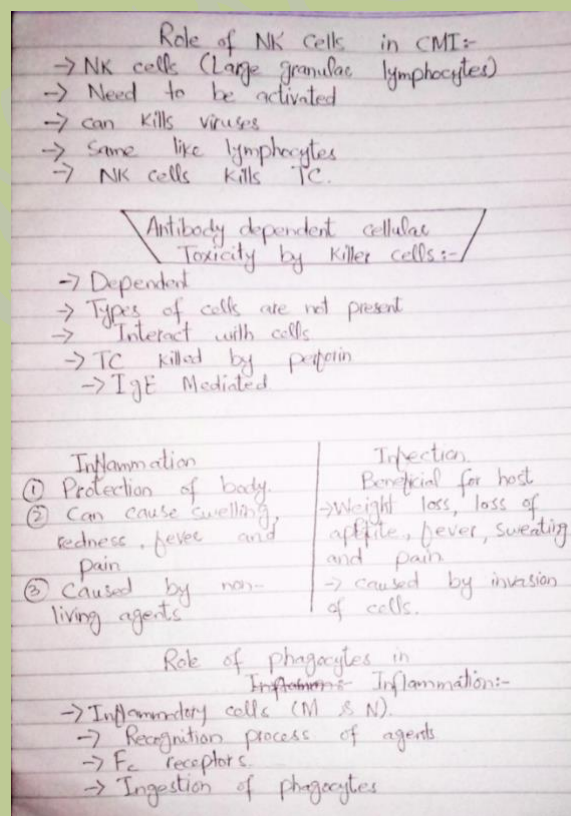
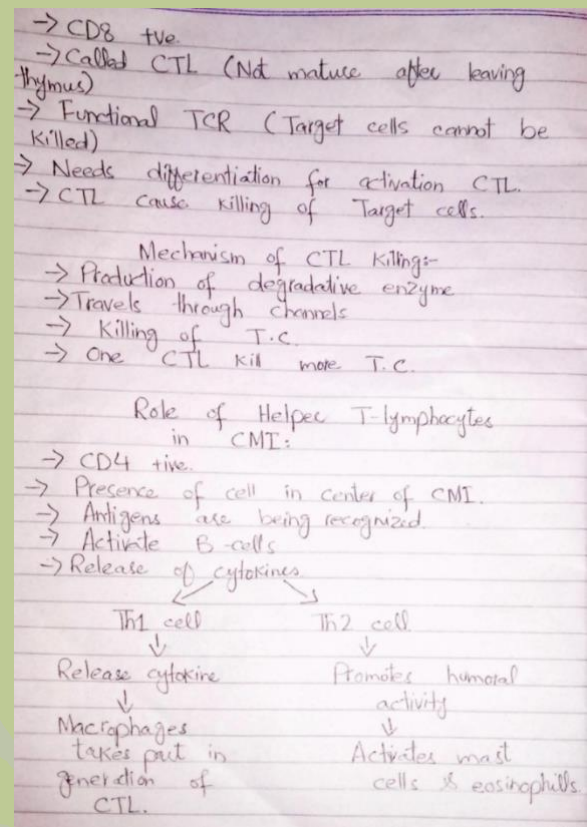
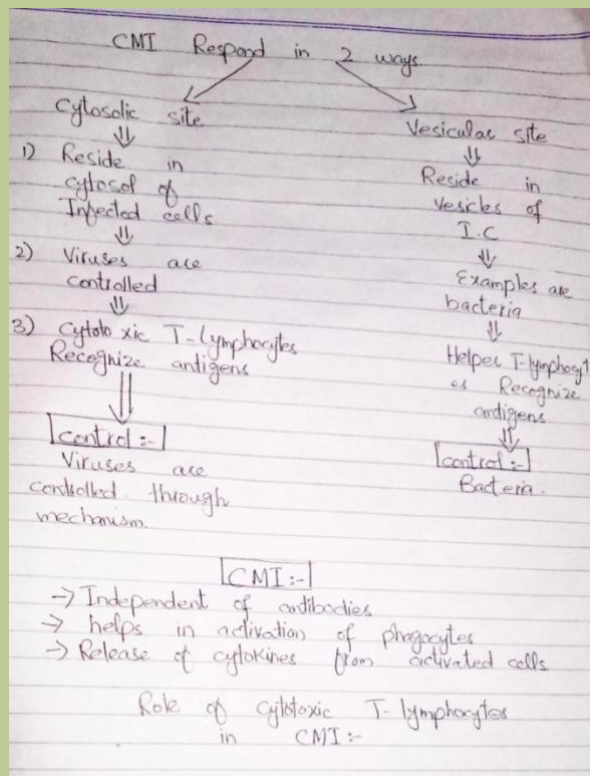
Mechanism:-
⇒ Escape from Immune system caused differently by different microbes.

Antigenic Variation:-
→ Organism mutate (antigenic surface).
→ Antibodies would not be able to work for long term (for protection).
→ Results in new strains
→ Influenza flu virus
→ E. coli, Neisseria gonorrhoeae.

Inhibition of complement activations
→ Degradation of C. proteins
→ Deviation cause complement deviation.
→ Resistance to insertion of membrane attack complex.

Resistance to phagocytosis:-
→ Inhibition of phagocytosis process
→ Trapping of bacteria e.g. S. Aureus.
→ Killing of phagocytes by toxins

Immune Response against Intracellular pathogens:-
→ Secretory immune molecules are ineffective against intracellular pathogens.
→ CMIR is primary defense.
→ T-lymphocytes play role in CMI.



BT302 IMMUNOLOGY
MID TERM SHORT NOTES
PREPARED BY BELLA

Differences B/W Monoclonal & Polyclonal Antibodies?

Monoclonal Antibodies

The antibodies which are produced from a single antibody producing B-cell are called as monoclonal antibodies. These antibodies also able to bind with single and unique antigen binding site (epitope). Monoclonal antibodies mainly consist of single subtype of IgG for example IgG1, IgG2 & IgG3.

Polyclonal Antibodies

The collection of antibodies from a different antibody producing B lymphocytes are called as polyclonal antibodies. These antibodies also able to bind with multiple epitopes on a same antigen. These antibodies are obtained from serum which has antibodies having different affinities. Polyclonal antibodies mainly belong to IgG class.

Write the phases of generation of Polyclonal Antibodies?

Phase I:

Antigen preparation (2 - 3 weeks)

- Peptide design
- Peptide synthesis
- Peptide conjugation

Phase II:

Antigen Immunization (8 - 10 weeks)

- 1 Antigen / 2 Rabbits
- ELISA detection
- 6 - 8 rounds of immunization

Phase III:

Antibody Purification (1 - 2 weeks)

- Antigen affinity purification
- Special affinity purification
- ELISA detection

Phase IV:

Antibody Validation (60 - 70 % success rate)

- Antibody delivery
- Antibody Validity
- Support and service

Write the production of Monoclonal Antibodies?

The production of monoclonal antibodies is done using immortal clone of cell with single antibody specificity. The immortal cells are those cells which don't have the ability to proliferate indefinitely. For this purpose, the normal antibody producing cell is being fused with an appropriate B-cell tumor line which is called as hybrid cell. Such kind of hybrid cells are for production of large number of antibodies.

Define hybrid cell line? also write its functions.

Hybrid cell line

The fusion of lymphoid tumor cell with normal B-Lymphocyte with single specificity hybrid cell line is called as hybrid cell line.

Function of hybrid cell line:

Hybrid cell line has dual function

- Immortality of tumor cell
- The production of antibody with single specificity

Write the principle/formation of hybridoma?

Principle of hybridoma:

Animal (mouse) is being injected with a specific antigen for a specific time point. After that, B-lymphocytes are isolated and sorted from the spleen of mouse. In the next phase, the B-Lymphocytes are fused with lymphocyte immortal cell which are myeloma cell. These cells provide the immortality to fused B-cells. The fusion between these two cells leads towards the generation of fused cell called as hybridoma. After fusion, the selection Hybridoma cell is usually done by using HAT medium (Hypoxanthine, aminopterin, thymidine) which allows the growth of hybridoma cells. While unfused cells undergo programmed cell death (apoptosis). The

selected hybridoma cells are harvested for production of monoclonal antibodies. For large number of monoclonal antibodies, hybridoma cells are propagated.

Write the usage of Monoclonal Antibodies?

Monoclonal antibodies have very significant applications in clinical and basic medical sciences which are as follows

1. Usage in prevention, diagnosis & treatment of disease
2. Immunophenotyping of Immune cells, Monoclonal antibodies are used against cell surface
3. molecules like Cluster of Differentiation (CD)
4. Typing of various tumors e.g., Leukemia
5. Immunohistochemistry of solid tumors
6. Use in Immunotherapy: Against various tumor marker e.g., CD20 B-Cell lymphoma
7. Use in Fluorescence Activated cell sorting (FACS)

Describe the role of Adjuvants in Polyclonal Antibodies Production?

Adjuvants are the substances which potentiates immune response against antigens. These can modulate towards desired immune response. However, with adjuvants there are mostly undesirable effects like cellular toxicity. That's why the selection of adjuvant is being made to get maximum immunostimulation. To produce polyclonal antibodies following adjuvants are used

1. Freund's Complete Adjuvants
2. Freund's Incomplete Adjuvants

Differentiate between Freund's Complete and Incomplete Adjuvants?

Freund's Complete Adjuvants (FCA)

Freund's complete adjuvant (FCA) are water-in-oil emulsion containing antigens with heat killed Mycobacterium tuberculosis. These adjuvants could stimulate both humoral & cell mediated immune response for antibody production.

Freund's Incomplete Adjuvants

These adjuvants are also water-in-oil emulsion containing antigens without heat killed Mycobacterium tuberculosis. These are used as booster antigen dose. These are used by mixing equal parts of the antigens.

Define immunological synapses?

In cell mediated immunity, cell to cell interaction is being orchestrated by a structure called as

immunological synapses. This synapse is generated during interaction between antigen presenting cells (APC) and T lymphocytes. From T-cells the T-cell receptor (TCR) contributes for this synapse. While from antigen processing cells (APC), MHC with processed peptides contribute for this synapse.

Difference Between MHC Class 1 and 2?

MHC Class 1 :

1. MHC class 1 are a class of major histocompatibility complex molecules found on the surface of all nucleated cells in mammals.
2. MHC class 1 molecules are expressed on all types of nucleated cells in the body.
3. MHC class 1 molecules are composed of three alpha domains and a single beta domain.
4. MHC class 1 molecules are composed of a single, membrane-spanning, alpha domain.
5. The three main types of MHC class 1 genes are MHC-A, MHC-B, and MHC-C.
6. The alpha domains are encoded on the MHC locus of chromosome 6 and beta chains are encoded on chromosome 15.
7. MHC class 1 molecules present endogenous antigens originated from the cytoplasm.
8. The alpha 1 and alpha 2 domains are involved in the presentation of antigens in MHC class 1 molecules.
9. MHC class 1 present antigens to cytotoxic T cells.
10. MHC class 1 molecules bind to the CD8+ receptors on the cytotoxic T cells.
11. MHC class 1 is responsible for the clearance of endogenous antigens.

MHC Class 2:

1. MHC class 2 are a class of major histocompatibility complex molecules mainly found on antigen presenting cells such as macrophages, dendritic cells, and B cells.
2. MHC class 2 molecules are expressed on the antigen presenting cells such as B cells, macrophages, and dendritic cells.
3. MHC class 2 molecules are composed of two alpha and beta domains.
4. MHC class 2 molecules are composed of two membrane-spanning alpha and beta domains.
5. The main type of MHC class 2 gene is MHC-D.
6. MHC class 2 genes are encoded on the chromosome 6.
7. MHC class 2 molecules present exogenous antigens originated extracellularly from foreign bodies such as pathogens.

8. The alpha 1 and beta 2 domains are involved in the antigen presentation in MHC class 2 molecules.
9. MHC class 2 present antigens to helper T cells.
10. MHC class 2 molecules bind to the CD4+ receptors on the helper T cells.
11. MHC class 2 is responsible for the clearance of exogenous antigens.

Write the structure of MHC Class 1 and 2 ?

Structure MHC Class 1 :

MHC class I molecules consist of one membrane-spanning α chain produced by MHC genes, and one β chain produced by the $\beta 2$ -microglobulin gene.

Structure MHC Class 2 :

MHC class II molecules consist of two membrane-spanning chains, α and β both produced by MHC genes.

What do you know about Antigen binding Groove of Class I MHC?

Antigen binding groove of class I MHC is composed of $\alpha 1$ & $\alpha 2$ domain. The antigenic peptides reside within the antigen binding groove. Within that groove peptide make contact with residue peptide that's why this region is highly polymorphic in nature. In this groove, up to 8-10 amino acids can accommodate.

Write some important Aspects of MHC?

Major Histocompatibility complex (MHC) have following significant aspects

- MHC molecules are membrane bounded
- These molecules are recognized by T-cells with antigen because of cell-to-cell interaction
- For an immune response, peptide must be bounded with MHC, this kind of immune response is called as one level control.
- Mature T-cell must have a TCR which recognize MHC bounded peptide, this kind of immune response is called as second level control.
- The expression of MHC is increased under the action of Cytokines
- Class I MHC recognize peptides from cytosol
- Class II MHC recognize peptides from vesicle
- The phenomenon of Polymorphism in MHC determines the specificity of immune response

Does MHC play important role in tissue transplantation?

Yes, MHC has very important role in tissue transplantation from one individual to other.

Define donor and recipient in tissue transplantation?

Donor :

The individual who donates the tissue is called as donor.

Recipient :

The individual who receives the tissue is called as recipient.

What do you know about HLA ?

The prospective donor & recipient are tested for compatibility prior to transplantation. That's why MHC is also called as Human leucocyte antigen (HLA). These molecules are the primary target of immune responses against allogeneic transplants. Moreover, HLA molecules are highly diverse in human population. HLA typing or tissue matching detects & classifies this diversity. There is extensive polymorphism in HLA that's why there are less chances of HLA matched donors for transplants. Approximately 25% of siblings inherit same HLA type.

What is B cell receptor (BCR) and how we recognized them?

B cell receptor (BCR) is composed of surface immunoglobulin molecule with a certain specificity for antigen. BCR can recognize following antigens in soluble form

- Proteins (Both Native & Denatured determinants)
- Nucleic acids
- Polysaccharides
- Some lipids
- Small molecules i.e haptens

What is B cell receptor (BCR) and how we recognized them?

T cell receptor (TCR) recognize mainly those antigens which are protein in nature. Those antigens are processed in fragmented form before recognition by TCR. Unlike BCR, TCR cannot recognize soluble form of antigens. The processed proteins (antigens) are recognized by TCR after association with a complex called as major histocompatibility complex (MHC). The MHC expressed on all nucleated cells of body. T-cells are grouped functionally on the basis of associated MHC with fragmented protein fragments

- Cytotoxic T-Cells which recognize antigens in association with class I MHC
- Helper T-Cells which recognize antigens in association with class II MHC

Define proteolysis?

All the processes that occur within the cell for fragmentation of antigens also called as proteolysis.

Define antigen processing?

The process in which the antigen is being processed is called as antigen processing. This process varies according to the source of antigen.

Write a note on Endogenous antigen?

The intracellular source of antigen which is also called as Endogenous antigen reside in the cytosol of antigen presenting cell like Viruses. For presenting such kind of endogenous antigens MHC I is required which is present in antigen presenting cells. MHC I express on all nucleated cells of body. Antigens (endogenous) are processed in a cellular organelle called as proteasome which is a complex having proteolytic activity. After proteolytic activity, the fragmented proteins move across the membrane of endoplasmic reticulum (ER) using transporter membrane. Likewise the synthesis & assembly of MHC I complex occur inside ER. Within ER MHC I form a stable complex with fragmented peptide & express on the surface of cell membrane for presenting to T-cells.

Difference between endogenous and exogenous antigens?

Endogenous antigens:

Endogenous antigens are antigens found within the cytosol of human cells such as viral proteins, proteins from intracellular bacteria, and tumor antigens.

Exogenous antigens:

Exogenous antigens are antigens that enter from outside the body, such as bacteria, fungi, protozoa, and free viruses.

Brief note on exogenous antigens?

The exogenous pathway is utilized by specialized antigen-presenting cells to present peptides derived from proteins that the cell has endocytosed. Proteins are endocytosed and degraded by acid-dependent proteases in endosomes; this process takes about an hour.

Short note on antigen presenting cell?

Antigen-presenting cells (APC) are cells that can process a protein antigen, break it into peptides, and present it in conjunction with class II MHC molecules on the cell surface where it may interact with appropriate T cell receptors.

Brief note on endogenous antigens?

Endogenous antigens are signals produced within your body's own cells having been infected by a virus that start an immune response. They alert cytotoxic T cells that a body cell is either infected with a virus, such as influenza, or has become cancerous.

What is antibody gene?

Antibody Genes Are Assembled from Separate Gene Segments During B Cell Development. Each Variable Region Is Encoded by more than one gene segment.

Write the Structure of Human Antibody Gene?

Antibody gene is divided into following two families based upon the structure of antibody

Light Chain Gene Family

This family encode for following light chains of antibody

- **Lambda Light Chains**

This gene is located on chromosome 22. Lambda light chain gene is composed of 4 constant (C) region genes. Among these everyone encodes for each subtype of Lambda. It also contains 30 Variable (V) genes. Each of V region gene composed of two exons, one L that encodes for Ladder exons and the other V that encodes for variable region. Upstream of each of the C genes there is, and additional exon called J (joining). The L, V, J and C exons are separated by introns.

- **Kappa Light Chains**

The kappa light chain gene family contains only one C region gene since there is only one type of kappa light chain. There are many V region genes (approximately 250) each of which has a leader exon and a V exon. In the κ gene family there are several J exons located between the V and C genes. All of the exons are separated by introns.

Heavy Chain Gene Family

Heavy chain gene is located at chromosome 14. It is composed of many constant C region gene for a each class & sub class. Each of C contain exons for hinge region & domains. There are approx.1000 variable genes. Each of V gene composed of L (Ladder) exons. Also contain V (variable) exons. In contrast to light chain gene family, it also carries additional D (diversity) exons which determines the diversity of antibody. Moreover, J (Joining) exons also exist. There are also introns in between exons.

Define somatic Hypermutation?

Somatic hypermutation is a process in which point mutations accumulate in the antibody V-regions of both the heavy and light chains is called as Somatic hyper mutation.

Write the role of Somatic Hyper mutation in diversity of antibodies?

Somatic hyper mutation has very significant role in the diversity of antibody. Once antigen enters the system, it stimulates a somatic mutational mechanism that generates antibodies of higher affinity and selects for the expression of those antibodies to produce a more effective immune response.

What is severe combined immunodeficiency?

Severe combined immunodeficiency (SCID) is a group of rare disorders caused by mutations in different genes involved in the development and function of infection-fighting immune cells. Infants with SCID appear healthy at birth but are highly susceptible to severe infections.

What is V (D) J combinational Diversity?

The component of antibody diversity that is generated by joining of various antibody gene segments. As in case of light chain genes, the V & J region genes combination occurs, while for heavy chain, all V, D & J region genes combination occurs. The D region combines first with J region and then V to form a complete heavy chain.

Write a note on regulation of V (D) J Recombination?

For the regulation of V (D) J recombination, there is some regions exist called as recombination signal sequence (RSS). These are flanked between V, D & J exons. These regions are composed of conserved regions in the form of haptamer & nonamer (based on number of base pairs) separated by two & one turn signals. The process recombination occurs between two & one turn Recombination occurs after the removal of introns between V & J in case of light chain like lambda & kappa chains. However, it occurs after the removal of introns between, D & J in case of heavy chain. Furthermore, there is involvement of certain enzymes which are called as recombination associated genes (RAG) which are responsible for this recombination. There are two types of RAG, RGA1 & RAG2 respectively. Clinically, the absence of RAG enzymes leads towards an immunodeficiency state called as severe combined immunodeficiency (SCID).

Define Somatic recombination?

Somatic recombination is a type of gene rearrangement by which cells of the adaptive immune system physically cut out small regions of DNA and then paste the remaining pieces of DNA back together in an error-prone way.

Define complement system/cascade?

The complement system also known as complement cascade is a part of the immune system that enhances the ability of antibodies and phagocytic cells to clear microbes and damaged cells from an organism to promote inflammation and attack the pathogen's cell membrane.

Explain Activation of the Complement System?

Complement proteins are activated through a series of cascading reactions which are called as complement activation. Followings are the important pathways for complement activation

The Classical Pathway

The classical pathway is activated when a complement protein called C1q binds either directly to a pathogen, or onto an antigen-antibody complex. This will then trigger cleavage of the subsequent complement proteins in the cascade, resulting in production of C3 convertase and its downstream effects. Its involvement in antigen-antibody complexes means it has a role in the adaptive immune response as well as the innate.

The Mannose-Binding Lectin (MBL) Pathway

Mannose-Binding Lectin (MBL) is a protein produced in the liver. Its role is to detect carbohydrates containing mannose on the surface of pathogens, activating a protease called MASP. MASP is responsible for cleaving complement components, which activates a similar cascade to the classical pathway, eventually producing C3 convertase.

The Alternative Pathway

The alternative pathway is usually activated by bacterial endotoxin, a lipopolysaccharide present on the outer membrane of gram-negative bacteria. This results in spontaneous hydrolysis of C3 into small amounts of factor C3b, which combines with other factors to produce C3 convertase.

Define Membrane attack complex/Lytic Pathway?

The membrane attack complex (MAC) or terminal complement complex (TCC) is a complex of proteins typically formed on the surface of pathogen cell membranes as a result of the activation of the host's complement system, and as such is an effector of the immune system.

Explain functions of Complement Proteins?

Complement system provides specific as well as non-specific resistance against infections. This kind of resistance is exhibited with following two important functions

Primary function

This function of complement proteins is considered primary as these activated proteins used to kill or lyse infectious agents like bacteria. For this complement proteins get deposited on the surface of infectious agents which make hole in those cells for their lysis or killing.

Secondary or Effector functions

These functions of complement proteins effect on other functions of immune system. These functions of complement proteins are as follows

- **Opsonization**

It is the process by which phagocytosis of microbes by immune cells get enhanced. Those proteins which undertake this process are called as opsonins. For example, some complement proteins like C3b, iC3b and C4b are good opsonins.

- **Chemotaxis**

The phenomenon by which phagocytic cells move towards the site of infection is called as chemotaxis, and the molecules involved in this process are called as chemotactic factors. The complement proteins are considered as good chemotactic factors like C5a.

- **Anaphylaxis**

Anaphylaxis is the form of immune response in which exaggerated response is generated against allergen. The molecules which produced such response are called as anaphylatoxins. The complement proteins like C4a, C3a & C5a act as potent anaphylatoxins.

Write note on Biological Active Products of Complement?

The complement system provides specific & non-specific resistance against infections. As a result of complement activation, there is production of various biological active molecules which are involved in resistance, anaphylaxis & inflammation. Followings are the important biological active products of complement.

Kinin production

C2b which is produced in classical pathway is called as Pro-Kinin. This Pro-Kinin is activated by serum factor called as Plasmin into Kinin which is a potent biological active compound. The excess C2b production causes undesirable effects like smooth muscles contraction and vasodilatation.

Anaphylatoxins

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Chemotactic factors

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Opsonins

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What are the role of CD40 in B-Cell Activation?

The CD40 activates both memory and naïve B cells and triggers their T cell stimulatory capacity.

Differentiate between Infectious diseases and Non-infectious diseases ?

Infectious diseases

An infectious disease can be transmitted from one person to another and are called communicable diseases. They are caused by microorganisms called pathogens such as bacteria, protozoans, viruses, etc. Examples: Cholera, chickenpox, malaria.

Non-infectious diseases

A non-infectious disease cannot be spread through person-to-person contact and they restrain within a person who has contracted them. These diseases are caused by pathogens, but other factors such as age, nutritional deficiency, the gender of an individual, and lifestyle also influence the disease. Examples: Diabetes, cancer, asthma, etc.

Write the Role of Immune System?

The immune system has a vital role that the most important function of immune system is to protect the body from the attack of infectious or invading microbes that could cause disease. The main role of the body's immune system is

- To fight disease-causing pathogens like bacteria, viruses, parasites, or fungi and to remove them from the body.
- To recognize and neutralize harmful substances from the environment.
- To fight disease-causing changes in the body, such as cancer cells.

Difference between Innate and Adaptive Immunity?

Innate Immunity

- The innate immunity is also known as the natural immunity.
- It is non-specific and is the first line of defense against pathogens.
- The innate immune system is something that already exists in the body.
- It is not inducible.
- Innate immunity fights for the any outsider invaders.
- The innate immune system responds quickly to pathogens.
- It has low potency and less effective as compared to the adaptive immunity. It is always present in the body.
- It is always present in the body.
- It has no memory of pathogens.
- **Example:** If skin cuts, if they cause swelling and inflammation, are an example of innate immunity at work.

Adaptive Immunity

- The adaptive immunity is also known as acquired immunity. It is a type of immune system that develops as we become more susceptible to infections or are vaccinated.
- Adaptive immunities are created because of exposure to external substances.
- It is specific and is the second line of defense against pathogens.
- Adaptive immunity only fights for certain infections.
- It is inducible.
- It only develops when foreign invaders enter the body.

- It has high potency and is very effective.
- The adaptive immune response is slower (1-2 weeks) than the innate immune system.
- Adaptive immune function is not present by birth, but it is developed throughout life.
- It has a long term memory of past viruses or pathogens that has been already encountered.
- **Example** : Vaccination of chickenpox so that we do not get chicken because the adaptive immunity remembers the foreign invaders.

What is a detoxification reaction?

The metabolic process by which toxins are changed into less toxic or more readily excretable substances in the act of detoxifying.

What is oxygen-dependent killing?

The oxygen compounds are toxic to both the invader and the cell itself, so they are kept in compartments inside the cell. This method of killing invading microbes by using the reactive oxygen-containing molecules is referred to as oxygen-dependent intracellular killing.

Define Oxygen-independent intracellular killing ?

Bacteria can be killed by pre-formed substances released from granules or lysosomes when they fuse with the phagosome. This is referred to as oxygen-independent intracellular killing. During phagocytosis glucose is metabolized via the pentose monophosphate shunt and NADPH is formed.

Explain Effector Functions of Complement Proteins?

Complement proteins have following effector functions for providing immunity against infectious agents

- **Opsonization (Opsonins)**

Various complement proteins bind with the surface of pathogens which facilitate the process of phagocytosis by phagocytes. This enhanced process of phagocytosis is called as opsonization, while complement proteins involved in this process are called as opsonins.

- **Chemoattraction (Chemoattractants)**

Complement proteins also served as chemoattractants for the movement of phagocytic cells towards the site of infection. The chemoattractant function of complement proteins increased the defense mechanism against invading agents.

- **Anaphylaxis (Anaphylatoxins)**

Complement proteins also involved in hypersensitivity (allergic) reactions which are also called as anaphylactic reactions in the form of exaggerated immune response. Such kind of

complement proteins are called as anaphylatoxins.

- **Pro-Inflammatory**

The role of complement proteins in inflammatory process is well established. The tissue response against infectious agent is termed as inflammatory response and the complement proteins contributing towards such kind of response are considered as pro-inflammatory.

What are the pathology related to Complement Proteins?

Complement proteins are considered as major humoral component of innate immune system. Once these proteins are activated lead the host towards following pathological conditions

- Acute Inflammatory response: Sudden inflammatory response is generated
- Capillary dilatation: Due to increase blood flow, there is capillary dilatation
- Exudation of plasma proteins & fluids (Edema): Accumulation of fluid in tissue spaces leading towards edema.
- Bronchoconstriction: there is constriction of bronchioles leading towards respiratory distress.
- Mast cells degranulation (Allergic reaction): Degranulation of mast cells leads towards the secretion of chemical mediators which finally become the cause of allergic reaction.

Write a note on Immune system disorders?

Immunodeficiencies

These arise when one or more parts of the immune system do not function. Immunodeficiencies can be caused in several ways, including age, obesity, alcoholism and AIDS. In some cases, immunodeficiencies can be inherited, for instance, in chronic granulomatous disease where phagocytes do not function properly.

Autoimmunity

In autoimmune conditions, the immune system mistakenly targets healthy cells, rather than foreign pathogens or faulty cells. In this scenario, they cannot distinguish self from non-self. Autoimmune diseases include celiac disease, type 1 diabetes, rheumatoid arthritis, and Graves' disease.

Hypersensitivity

With hypersensitivity, the immune system overreacts in a way that damages healthy tissue. An example is anaphylactic shock.

What are Phagocytes and its types?

These cells surround and absorb pathogens and break them down, effectively eating them. There are several types, including:

- **Neutrophils** — these are the most common type of phagocyte and tend to attack bacteria.
- **Monocytes** — these are the largest type and have several roles.
- **Macrophages** — these patrol for pathogens and also remove dead and dying cells.
- **Mast cells** — they have many jobs, including helping to heal wounds and defend against pathogens.

Difference between Primary and Secondary lymphatic tissues?

Primary lymphatic Tissues

Primary lymphatic organs are where lymphocytes are formed and mature. They provide an environment for stem cells to divide and mature into B- and T- cells. There are two primary lymphatic organs: the red bone marrow and the thymus gland. Both T-cell and B-cells are 'born' in the bone marrow. However, whereas B cells also mature in the bone marrow, T-cells have to migrate to the thymus, which is where they mature in the thymus.

Secondary lymphoid Tissues

Secondary lymphoid tissues are arranged as a series of filters monitoring the contents of the extracellular fluids, i.e., lymph, tissue fluid and blood. The lymphoid tissue filtering each of these fluids is arranged in different ways. Secondary lymphoid tissues are also where lymphocytes are activated. These include lymph nodes, tonsils, spleen, Peyer's patches and mucosa associated lymphoid tissue (MALT).

Difference between Lymphoid and Myeloid lineage?

Lymphoid lineage

The lymphoid lineage of immune cells give rise to various cells of specific immune systems like consists of T, B and natural killer (NK) cells.

Myeloid lineage

The myeloid lineage includes a few morphologically, phenotypically, and functionally distinct cell types such as different subclasses of granulocytes (neutrophils, eosinophils and basophils), monocytes–macrophages, erythrocytes, megakaryocytes and mast cells.

What is Lymphocyte Recirculation?

Lymphocyte recirculation refers to the continuous transport of large numbers of naïve lymphocytes between the blood and lymphatic systems. After naïve lymphocytes are produced in the thymus or bone marrow, they enter the blood, where they spend about 30 min, before migrating to the lymphoid and nonlymphoid organs.

Define Lymphocytes Migration & Localization?

Lymphocytes migrate from the lymph node into blood following an S1P gradient that is

maintained by the high levels of S1P in the blood and lymph, which far exceed those at tissues. Tissue levels of S1P are tightly controlled due to intracellular degradation by S1P lyase or dephosphorylation by S1P phosphatases. Some lymphocytes migrate to the thymus, where they mature into T cells; others remain in the bone marrow, where—in humans—they develop into B cells.

Write the principles of the clonal selection theory?

There are following four basic principles of the clonal selection theory

1. Each lymphocyte bears a single type of receptor with a unique specificity.
2. Interaction between a foreign molecule and a lymphocyte receptor capable of binding that molecule with a high affinity leads to lymphocyte activation.
3. The differentiated effector cells derived from an activated lymphocyte will bear receptors of an identical specificity to those of the parental cell from which that lymphocyte was derived.
4. Lymphocytes bearing receptors for self-molecules are deleted at an early stage in lymphoid cell development and are therefore absent from the repertoire of mature lymphocytes.

Define T-cell receptor? Also write its structure.

T-cell receptor

The T-cell receptor is a protein complex found on the surface of T cells, or T lymphocytes, that is responsible for recognizing fragments of antigen as peptides bound to major histocompatibility complex molecules.

Structure of T-cell receptor

T cell receptors are made up of two polypeptide chains that together compose one antigen binding region. Approximately 95% of TCRs are composed of an alpha and a beta chain, while the remaining 5% of TCRs are made up of gamma and delta chains. There are multiple types of T cells, and each has a specialized function.

Define B-cell receptor? Also write its structure.

B cell receptor

The B cell receptor (BCR) is a transmembrane protein on the surface of a B cell. A B cell receptor is composed of a membrane-bound immunoglobulin molecule and a signal transduction moiety. The BCR for an antigen is a significant sensor that is required for B cell activation, survival, and development.

B cell structure and function

B cell receptors are made up of four peptides – two light chains and two heavy chains that comprise two antigen-binding regions. Light chains are classified as either kappa or lambda,

while the heavy chains can be IgG, IgA, IgM, IgD, or IgE isotypes.

What is the Diversity of T-cell receptor?

T-cell receptor (TCR) diversity, a prerequisite for immune system recognition of foreign antigens, is generated in the first two decades of life in the thymus and then persists to an unknown extent through life via homeostatic proliferation of naïve T cells.

Write the CD3 complex Structure & Function?

The CD3 stands for Cluster of differentiation 3. This structure is present adjacent to TCR. This is basically the transducing element of TCR. It exists as a complex form which is composed of five following proteins that's why it is also called as CD3 complex.

1. One Gamma(γ)
2. One delta(δ)
3. Two Epsilon (ϵ)
4. Two Zeta (ζ)

How many signals are required for the activation of T-cells ?

Cell Surface Molecules involved in T-Cell for the activation of T-cells following two signals are required

1. Signal 1 is generated after engagement of TCR with Antigen/MHC complex.
2. Signal 2 is generated after engagement of supporting molecules with their corresponding ligands.

What is Helper and Cytotoxic T cells?

Helper T cells

A type of immune cell that stimulates killer T cells, macrophages, and B cells to make immune responses. A helper T cell is a type of white blood cell and a type of lymphocyte. Also called CD4-positive T lymphocyte.

Cytotoxic T cells

A cytotoxic T cell (left) recognizes antigens on the surface of a cell infected with a virus (right), enabling the T cell to bind to and kill the infected cell. The cells are also known as CD8+ T cells.

Define Hematopoietic stem cells?

Hematopoietic stem cells (HSCs) are the stem cells that give rise to other blood cells. This process is called hematopoiesis. In adults, hematopoiesis occurs in the red bone marrow, in the core of most bones. The red bone marrow is derived from the layer of the embryo called the mesoderm.

Write the Function of CD3?

CD3 (cluster of differentiation 3) is a protein complex and T cell co-receptor that is involved in activating both the cytotoxic T cell (CD8+ naive T cells) and T helper cells (CD4+ naive T cells). It is composed of four distinct chains.

Difference between B cell dependent and B cell independent t cell activation?

B cell dependent t cell activation

T cell-dependent activation of B cells involves processing and presentation of protein antigens to helper T cells, activation of the B cells by cytokines secreted from activated TH2 cells, and plasma cells that produce different classes of antibodies because of class switching.

B cell independent t cell activation

T cell-independent activation of B cells involves cross-linkage of BCRs by repetitive nonprotein antigen epitopes. It is characterized by the production of IgM by plasma cells and does not produce memory B cells.

Difference between Single and double immunodiffusion?

Single immunodiffusion

Single Immunodiffusion is a technique used extensively for the quantitative estimation of antigens. Antibody of known specificity is distributed evenly in an agar gel and a sample containing the antigen of interest is placed in a well within the gel.

Double immunodiffusion

Double immunodiffusion is an agar gel immunodiffusion. It is a special precipitation reaction on gels where antibodies react with specific antigens forming large antigen-antibody complexes which can be observed as a line of the precipitate.

What is Major histocompatibility complex (MHC) ?

Major histocompatibility complex (MHC), group of genes that code for proteins found on the surfaces of cells that help the immune system recognize foreign substances. MHC proteins are found in all higher vertebrates. In human beings the complex is also called the human leukocyte antigen (HLA) system.

On what basis antibodies are classified?

Antibodies are classified according to heavy chain type, which is encoded by a gene on chromosome 14. The different classes are IgG, IgA, IgM, IgD and IgE; in descending order of abundance in serum.

How nature of membrane bound, and secreted antibodies is determined?

The difference between the two forms of IgM is entirely at the C-terminus. The μ s (secreted) chain has a hydrophilic peptide additional to the C μ 4 domain and the μ m (membrane-bound)

chain has a hydrophobic peptide replacing the hydrophilic C-terminal peptide of the μ s chain²⁻⁴.

Difference between Self antigen and Non-Self antigen?

Self-antigen

Self-antigens are by convention antigens in the body of an individual.

Non-Self antigen

Any substance that is recognized as foreign and can trigger an immune response is called an antigen.

Define Accessory Molecules? Also write down the important accessory molecules?

Accessory molecules are cell surface proteins that promote or suppress the response of immune cells. Accessory molecules include cell surface molecules involved in immune cell activation and differentiation, in inhibition of immune cell activity and in cell homing and adhesion. Followings are the important accessory molecules.

- **CD4**

This molecule of T-cell is for binding to class II MHC which ensures the binding of T-helper (Th) cells to antigen presenting cells

- **CD8**

This accessory molecule is for binding to Class I MHC which ensures the binding of cytotoxic T (Tc) cells to target cell

- **CD2**

This accessory molecule on the surface of T-cell has a ligand known as Leucocyte Function Antigen 3 (LFA-3) on antigen presenting cells (APC)

- **Leucocyte Function Antigen 1 (LFA-1)**

This molecule has ligand known as Intracellular Adhesion Molecule-1 (ICAM-1)

- **CD28**

CD28 is the accessory molecule on T-cell and has ligand called as B7-1/B7-2

Define Modulators of immune system?

A substance that stimulates or suppresses the immune system and may help the body fight cancer, infection, or other diseases. Specific immune system modulators, such as monoclonal antibodies, cytokines, and vaccines, affect specific parts of the immune system.

What are the antibodies? and also write its types ?

An antibody is a protein component of the immune system that circulates in the blood,

recognizes foreign substances like bacteria and viruses, and neutralizes them.

Types of Antibodies :

There are five types of antibodies

- **Immunoglobulin A (IgA):**

IgA is almost 15% present in the blood. It is found in the lining of the respiratory tract and digestive system even it is also present in the saliva, tears, and breast milk. IgA provides protection against infection in the mucosal areas of the body. There are two IgA phases of IgA: IgA1 and IgA2.

- **Immunoglobulin G (IgG):**

IgG is a very common antibody and about 70 to 80% is present in the blood and other body fluids. It protects against infections caused by bacteria and viruses. IgG may take some time to developed after infection or vaccination. Only IgG antibody can pass through the placenta. There are four categories of IgG: IgG1, IgG2, IgG3, and IgG4.

- **Immunoglobulin M (IgM):**

The IgM antibody is mainly found in blood and lymph fluid. IgM is the body's first antibody when it fights new infections. They multiply for a few weeks and then decline as IgG production begins.

- **Immunoglobulin E (IgE):**

IgE is a type of antibody produced by the immune system and is usually found in small amounts in the blood. There may be high levels when the body reacts to allergies or fights infections. Its molecular weight is 200,000 and 0.002%. And its serum concentration in the blood is 10 to 400 ng / ml.

- **Immunoglobulin D (IgD):**

IgD is a unique antibody with low concentration in serum and the exact activity of IgD is unknown. IgD introduces approximately 0.25% of blood and its molecular weight is 185 kDa while its half-life is 2.8 days. However, it makes up about 1% of the protein in the plasma membrane. IgD is a rarely antibody that has only a small amount in the blood.

What is cellular energy?

Cells do not use the energy from oxidation reactions as soon as it is released. Instead, they convert it into small, energy-rich molecules such as ATP and nicotinamide adenine dinucleotide (NADH), which can be used throughout the cell to power metabolism and construct new cellular components.

Define antigen ?

An antigen is a molecule or molecular structure that can bind to a specific antibody or T-cell receptor. The presence of antigens in the body may trigger an immune response.

What is class switching recombination?

Class switch recombination is a process by which proliferating B cells rearrange the constant region genes in the immunoglobulin heavy chain locus to switch from expressing one class of immunoglobulin (such as IgM) to another (such as IgG).

Define following terms ?

Immunity

It is the resistance to disease specifically infectious disease or infections.

Immune system

The collection of cells, tissues and molecules that mediate resistance to infection is called as immune system.

Immune response

The coordinated reaction of cells and molecules of immune system against infectious microbes is termed as immune response.

Immunology

The study of immune system & immune response with physiological function to prevent infections and to eradicate established infections inside body.

What is the response of phagocytes to infection?

Phagocytes are crucial in fighting infections, as well as in maintaining healthy tissues by removing dead and dying cells that have reached the end of their lifespan. During an infection, chemical signals attract phagocytes to places where the pathogen has invaded the body.

Write the note on Initiation of Phagocytosis?

Phagocytic cells have a variety of receptors on their cell membranes through which infectious agents bind to the cells. These include:

- **Fc receptors**

Bacteria with IgG antibody on their surface have the Fc region exposed and this part of the antibody molecule can bind to the receptor on phagocytes. Binding to the Fc receptor requires prior interaction of the antibody with an antigen. Binding of IgG-coated bacteria to Fc receptors results in enhanced phagocytosis and activation of the metabolic activity of phagocyte.

- **Complement receptors**

Phagocytic cells have a receptor for the 3rd component of complement, C3b. Binding of C3b coated bacteria to this receptor also results in enhanced phagocytosis and stimulation of the respiratory burst.

- **Scavenger receptors**

Scavenger receptors bind a wide variety of polyanions on bacterial surfaces resulting in phagocytosis of bacteria.

- **Toll-like receptors**

Phagocytes have a variety of Toll-like receptors (Pattern Recognition Receptors or PRRs) which recognize broad molecular patterns called PAMPs (pathogen associated molecular patterns) on infectious agents. Binding of infectious agents via Toll-like receptors results in phagocytosis and the release of inflammatory cytokines (IL-1, TNF-alpha and IL-6) by the phagocytes.

Write the Control of Intracellular Pathogens and Extracellular Pathogens?

Control of Intracellular Pathogens

Secretory molecules of immune system like antibodies are not able to provide effective immune response against those pathogens which attach the host cells by residing intracellularly. These pathogens are controlled by cell mediated immune response which is being provided by T-lymphocytes. These T-lymphocytes provide protection against intracellular pathogens according to the nature of pathogen location inside the cells. There are following two major kinds of T-lymphocytes which respond against intracellular pathogens according to the location of pathogen inside the cells

- Cytotoxic T-Lymphocytes
- Helper T-Lymphocytes

Control of Extracellular Pathogens

Immune System provides protection against diverse pathogens which are highly contagious in nature. This kind of defense against pathogen is based upon the foreign nature of pathogen. These pathogens can attack the host by residing outside of host cells. Such kind of pathogens are called as extracellular pathogens. Antibodies control the extracellular pathogens by the following three important ways

- Neutralization
- Opsonization
- Complement activation

Define following terms ?

Anamnestic response

It is a rapid production of an antibody on the second encounter with the same antigen. Memory response is also called as anamnestic response.

Specificity of antibody response

Specificity is the degree to which an immune response epitope between antigenic variants.

Nature of B-Cell Activation

B cells are activated when their B cell receptor (BCR) binds to either soluble or membrane bound antigen.

Write down the B-Cell Receptor & Co-stimulation for Maturation?

The process of activation of B-cells occur after cross linking of BCR through antigen. As the cross linking initiates, there is activation of a molecule called as Src for cellular signaling. There are following two biochemical processes are activated during antibody synthesis

- The Activation of Phospholipase C (PLC γ)
- The Phosphorylation of adapter protein called as SLP which in turn cause the activation of Ras & Rac proteins

Write down the Co-Stimulatory Molecules for Activation & Maturation of T-Cells?

The molecules required for activation of T-Cells after binding to its ligands on antigen presenting cells. The following two signals required for T-cells activation

- Engagement of TCR with antigen which is associated with MHC
- Engagement of co-stimulatory molecules with their ligands

What is class switching recombination?

Class switch recombination is a process by which proliferating B cells rearrange the constant region genes in the immunoglobulin heavy chain locus to switch from expressing one class of immunoglobulin (such as IgM) to another (such as IgG).

What is coagulation protein?

Coagulation is defined as the change in the structure of protein brought about by heat, mechanical action, or acids.

Define following terms?

Lactoferrin & Transferrin's

These proteins limit bacterial growth, by binding iron which is an essential nutrient for bacteria.

Interferons

Interferons are the secretory molecules which are proteins in nature. These have ability to restrict virus replication in cells.

Cytokines

These are secretory molecules which act as mediators between immune cells. IL-1 induces fever and the production of acute phase proteins, some of which are antimicrobial because they can opsonize bacteria.

